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Introduction of Machine Learning

Segment-Aware and Time-Aware Market Basket Analysis

FINAL PROJECT DEMO



GROUP I

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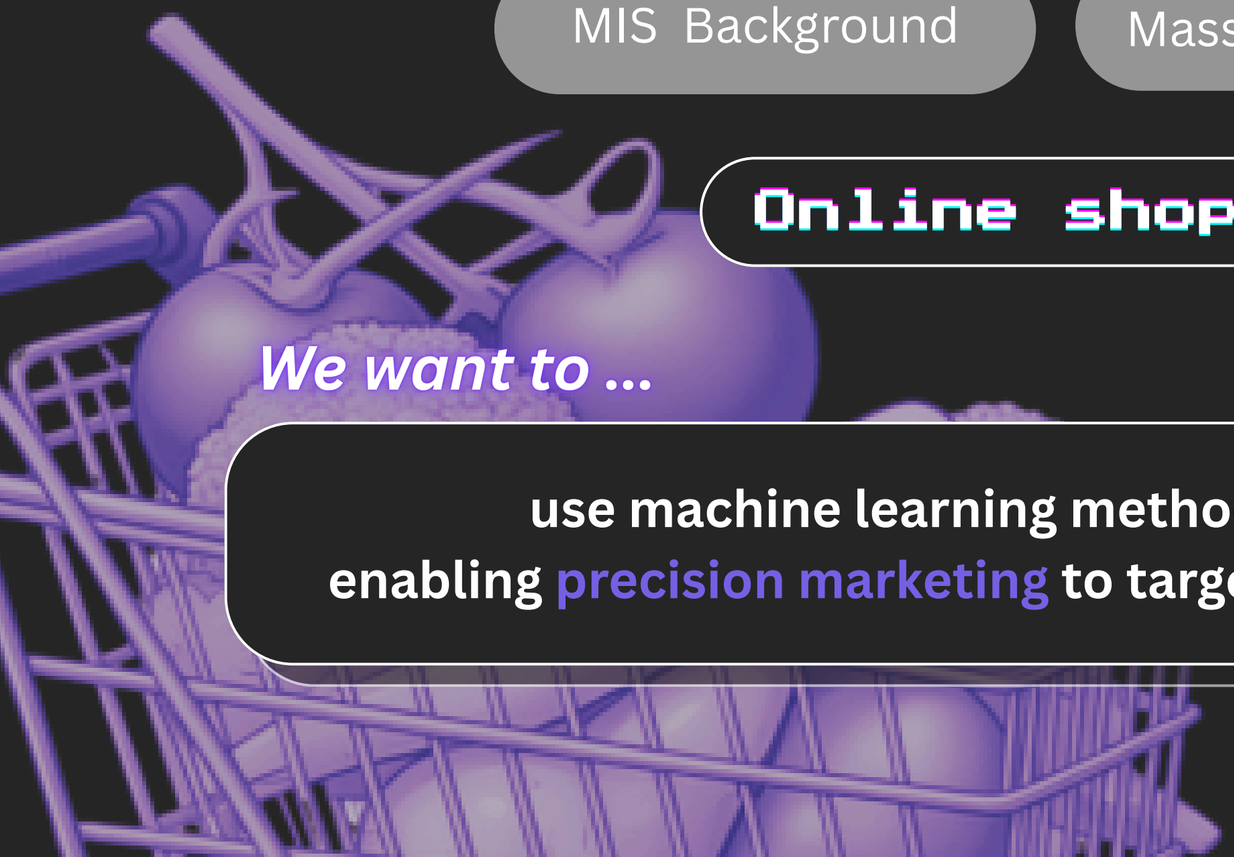
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1 MOTIVATION & GOAL

MIS Background

Massive & Diverse E-Commerce Market

Online shopping platform



We want to ...

use machine learning methods to solve inefficient marketing,
enabling precision marketing to target audiences and creating data into value.

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DATASET

- **Source:** Instacart Online Grocery Basket Analysis from **Kaggle**.
- **Complexity:** **Over 3 million orders** requiring the **joining of 7+ relational tables**.
- **Feature Construction:** Transforming **raw transaction logs** into user-level behavioral metrics, such as reorder rates.
- **Data Cleaning :** Handling **missing values** in order **intervals** and parsing product categories for analysis.

URL: *https://www.kaggle.com/datasets/yasserh/instacart-online-grocery-basket-analysis-dataset?select=orders.csv*

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TECHNIQUES, METHODS

3 CUSTOMER SEGMENTATION

Feature Engineering（RFM + behavioral）

Convert transaction data into user-level features (e.g., frequency, total items, reorder rate) to represent each customer for clustering.

K-means

There are no predefined labels, we use K-means to cluster customers based on behavioral features. It is simple, efficient, and easy to interpret.





TECHNIQUES, METHODS

MARKET BASKET ANALYSIS

Apriori / FP-growth

Use association rule mining to discover frequently co-purchased products.

Analyze patterns within each customer segment.

Support, Confidence and Lift:

Support

- Frequency of itemset in all transactions

$$Support(A, B) = \frac{\#\text{ transactions containing A and B}}{\text{total transactions}}$$

Confidence

- Probability of buying B given A

$$Confidence(A \rightarrow B) = \frac{Support(A, B)}{Support(A)}$$

Lift

- Strength of association between A and B

$$Lift(A \rightarrow B) = \frac{Confidence(A \rightarrow B)}{Support(B)}$$

TECHNIQUES, METHODS

3 TIME-AWARE BASKET ANALYSIS

Our Key Feature!!

Apriori / FP-growth

Apply association rule mining under different time conditions to discover time-dependent purchasing patterns.

Support, Confidence and Lift:

Compute support, confidence, and lift within each time period to evaluate time-dependent associations.



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EXPECT EXPERIMENTAL RESULTS

- Customer Segmentation
- Basket Analysis
- Time-aware Analysis

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EXPECT EXPERIMENTAL RESULTS

Customer Segmentation

- Behavioral Features
- Behavioral and Product Preference Features
- Behavioral and Product Preference Features (Without Produce and Dairy Eggs)
- Customer Behavioral and Product Preference Features

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Behavioral Features

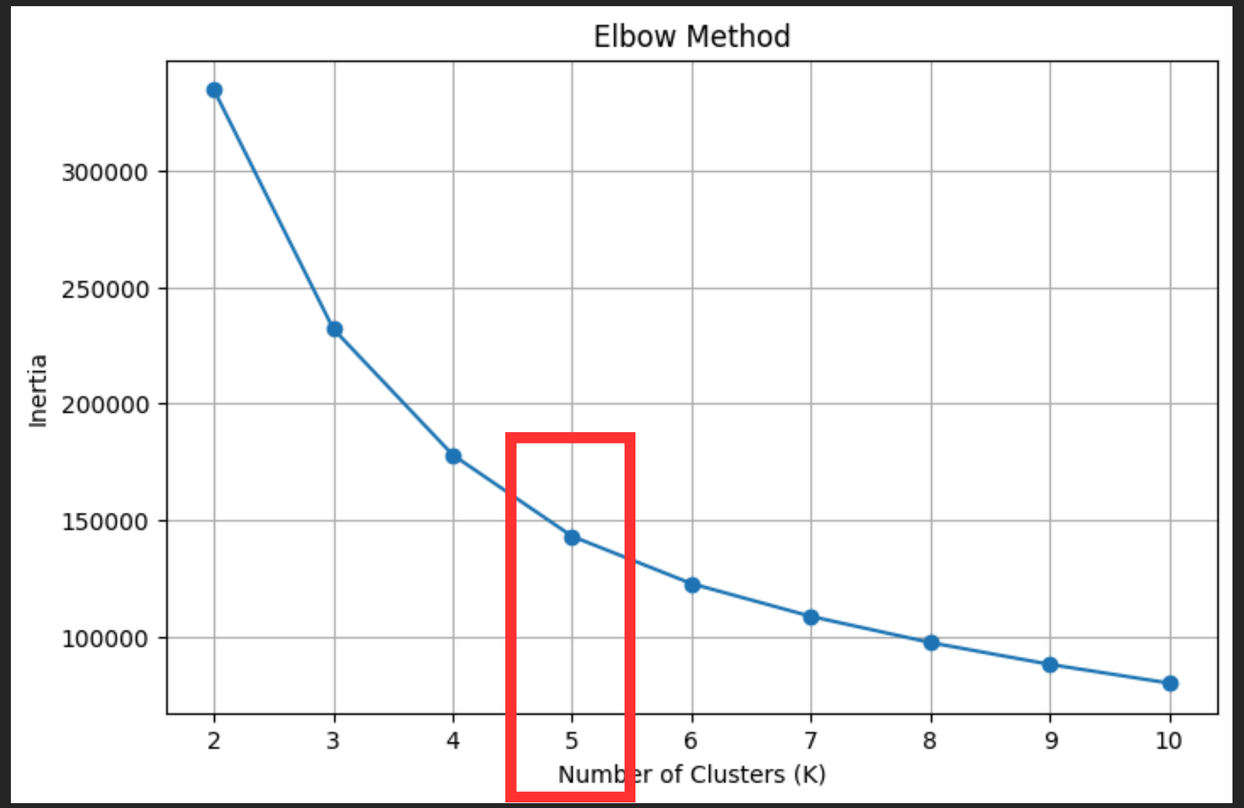


Figure 1. Elbow Method for Behavioral Feature Clustering

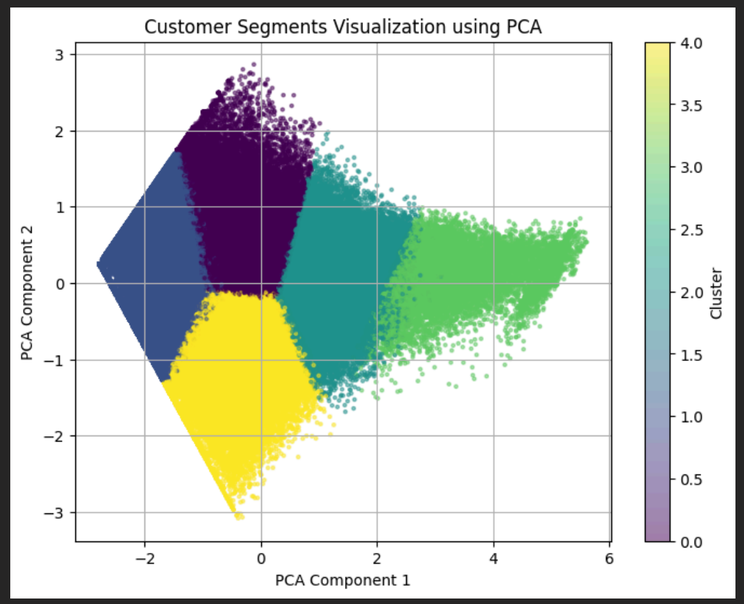


Figure 2. PCA Visualization of Behavioral Feature Clusters

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Behavioral Features

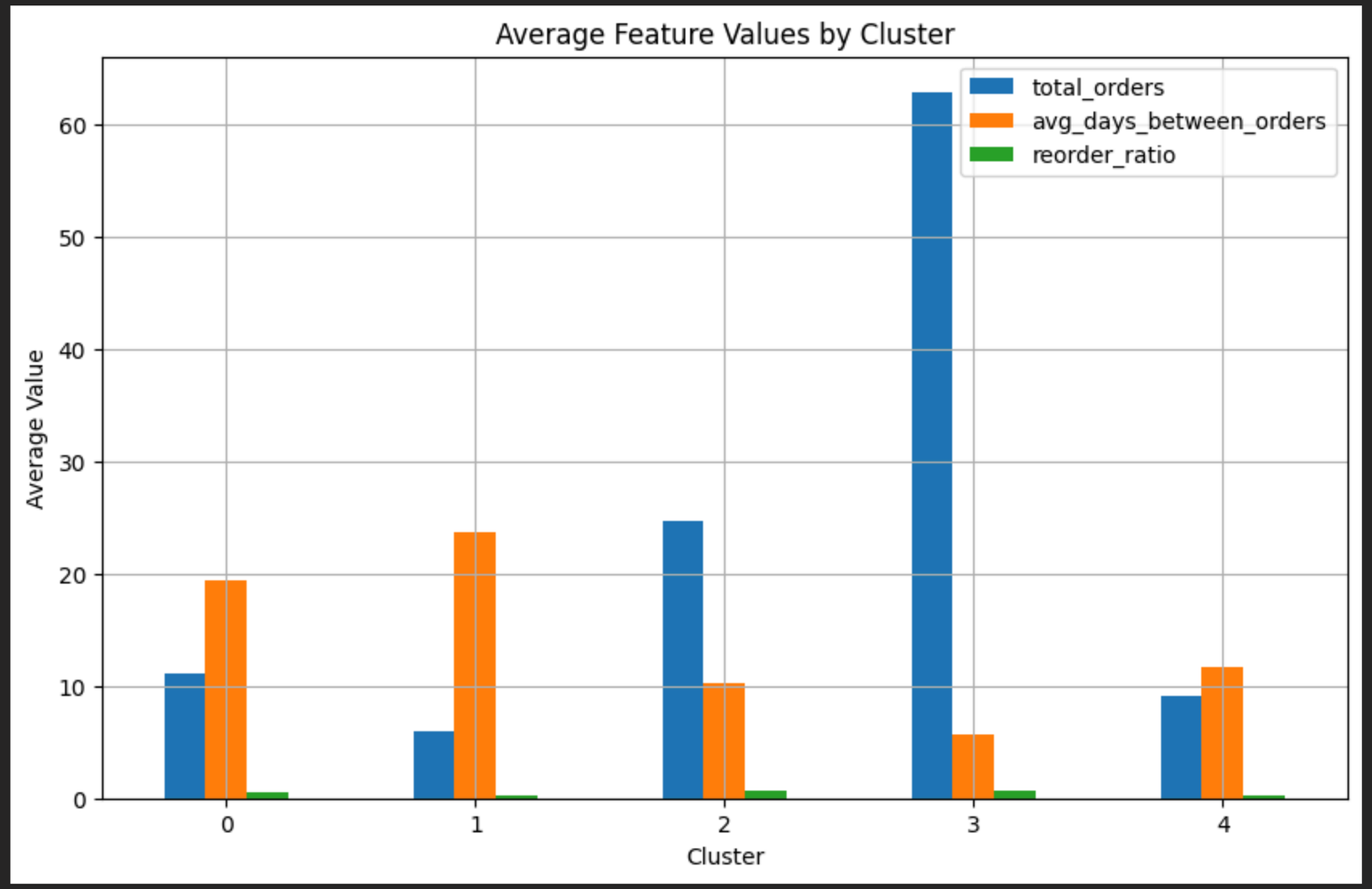


Figure 3. Average Behavioral Feature Values by Cluster

Behavioral Features

Based on the K-means clustering results, we categorized customers into the following groups:

Highly Loyal Customers: (Cluster 3)

- High total_orders, reorder_ratio and low avg_days_between_orders.
- Highly loyal and active customers with frequent purchases and high reorder rates.

Stable Customers: (Cluster 2)

- Moderate-to-high total_orders, relatively high reorder_ratio, and medium purchasing intervals.
- Stable purchasing behavior and consistent shopping habits.

Moderate / Casual Customers: (Cluster 0, Cluster 4)

- Moderate total_orders and reorder_ratio, with less frequent purchasing behavior.
- Purchase occasionally but show lower customer loyalty.

Inactive Customers: (Cluster 1)

- Low total_orders, low reorder_ratio, and high avg_days_between_orders.
- Inactive customers with infrequent purchases and low engagement.

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Behavioral Features

Table 1. Behavioral Characteristics of Customer Clusters

Cluster	total_orders	avg_days_between_orders	reorder_ratio
0	11.0458	19.3742	0.5064
1	6.0143	23.7489	0.2116
2	24.6770	10.2112	0.6252
3	62.9285	5.7423	0.7347
4	9.1345	11.7486	0.2842

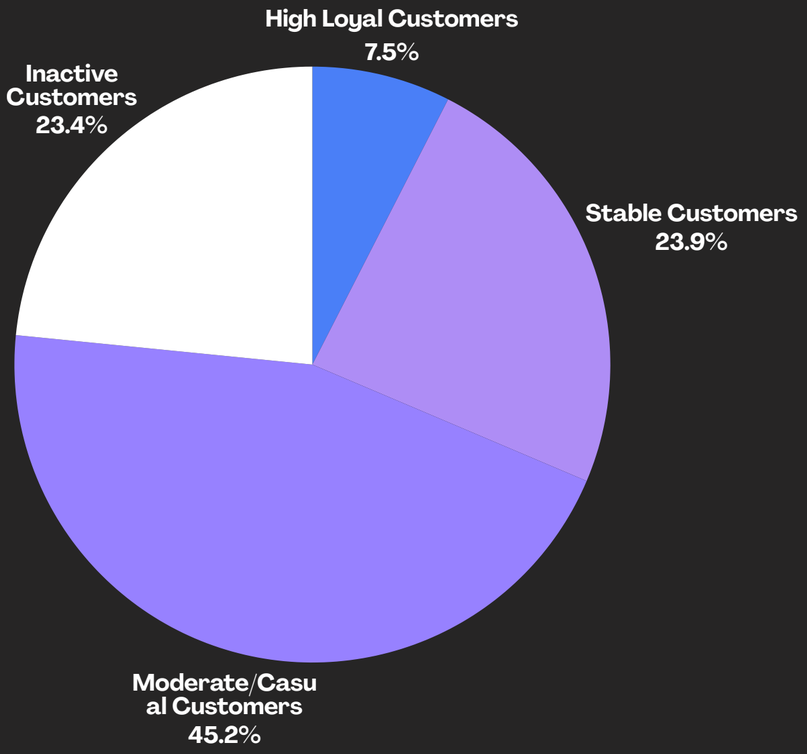


Figure 4. Proportion of Customer Behavioral Segments

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Behavioral Features

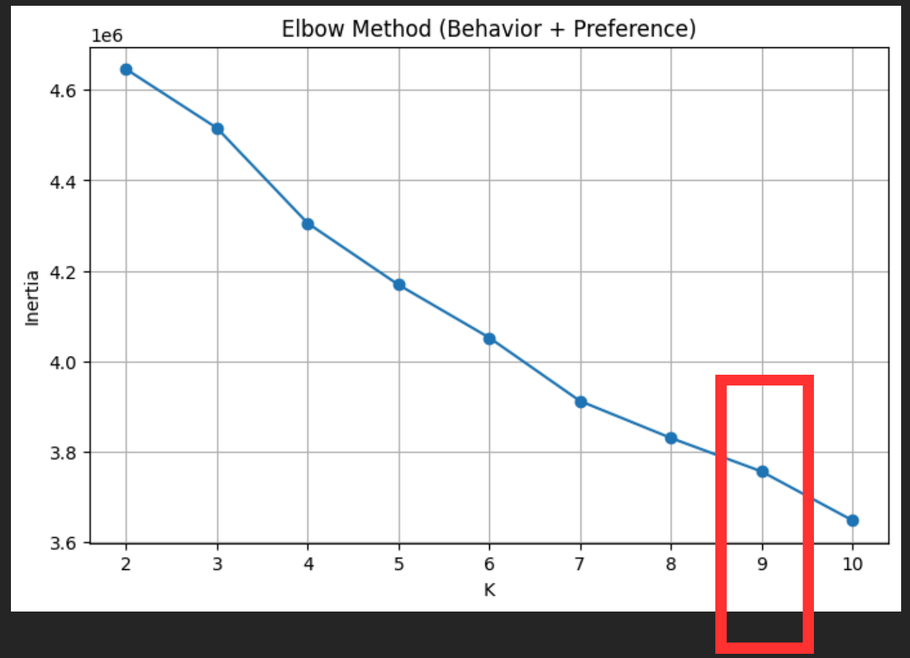


Figure 5. Elbow Method for Behavioral and Product Preference Feature Clustering



Figure 6. PCA Visualization of Behavioral and Product Preference Clusters

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Behavioral and Product Preference Features

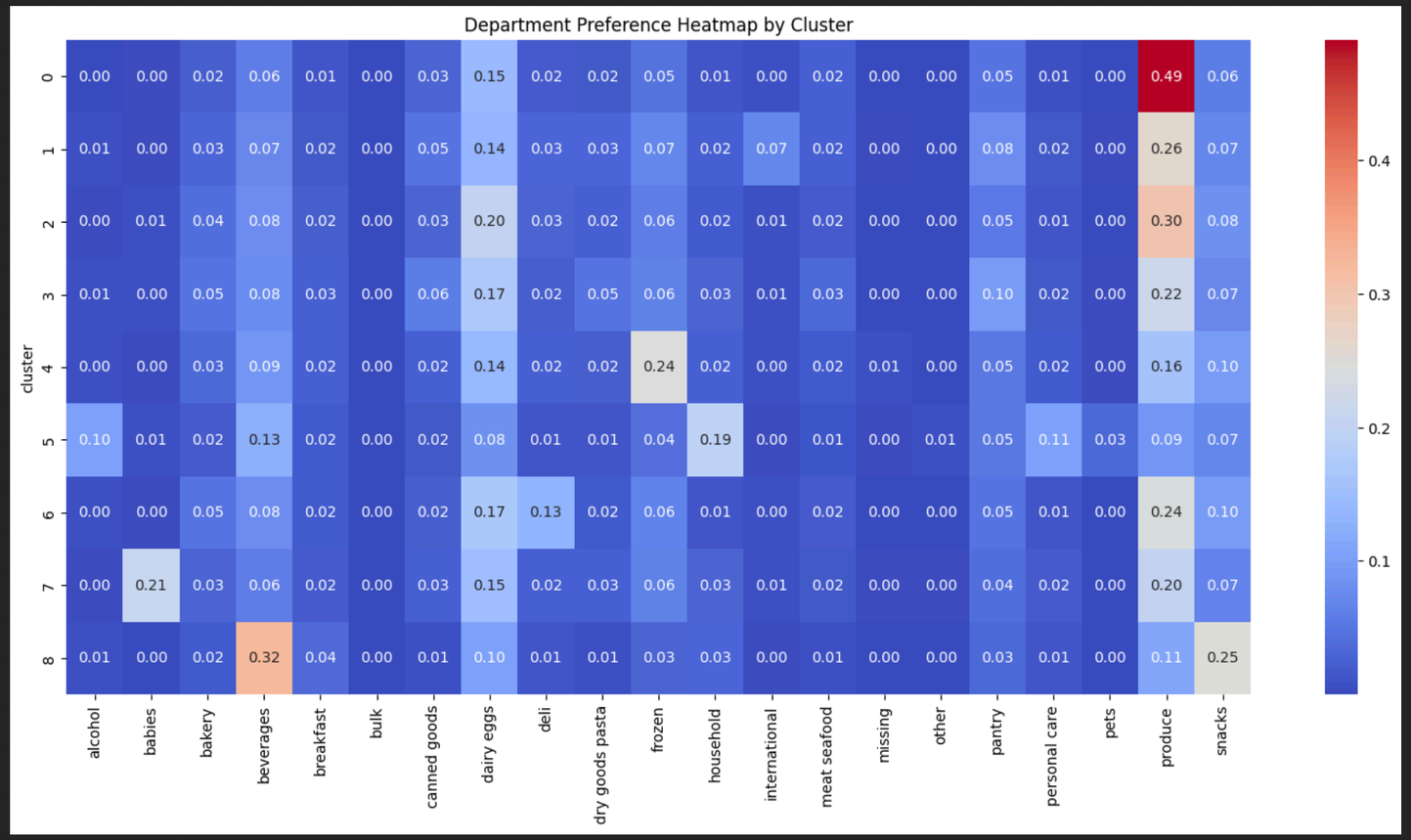


Figure 7. Department Preference Heatmap by Cluster

Behavioral and Product Preference Features©

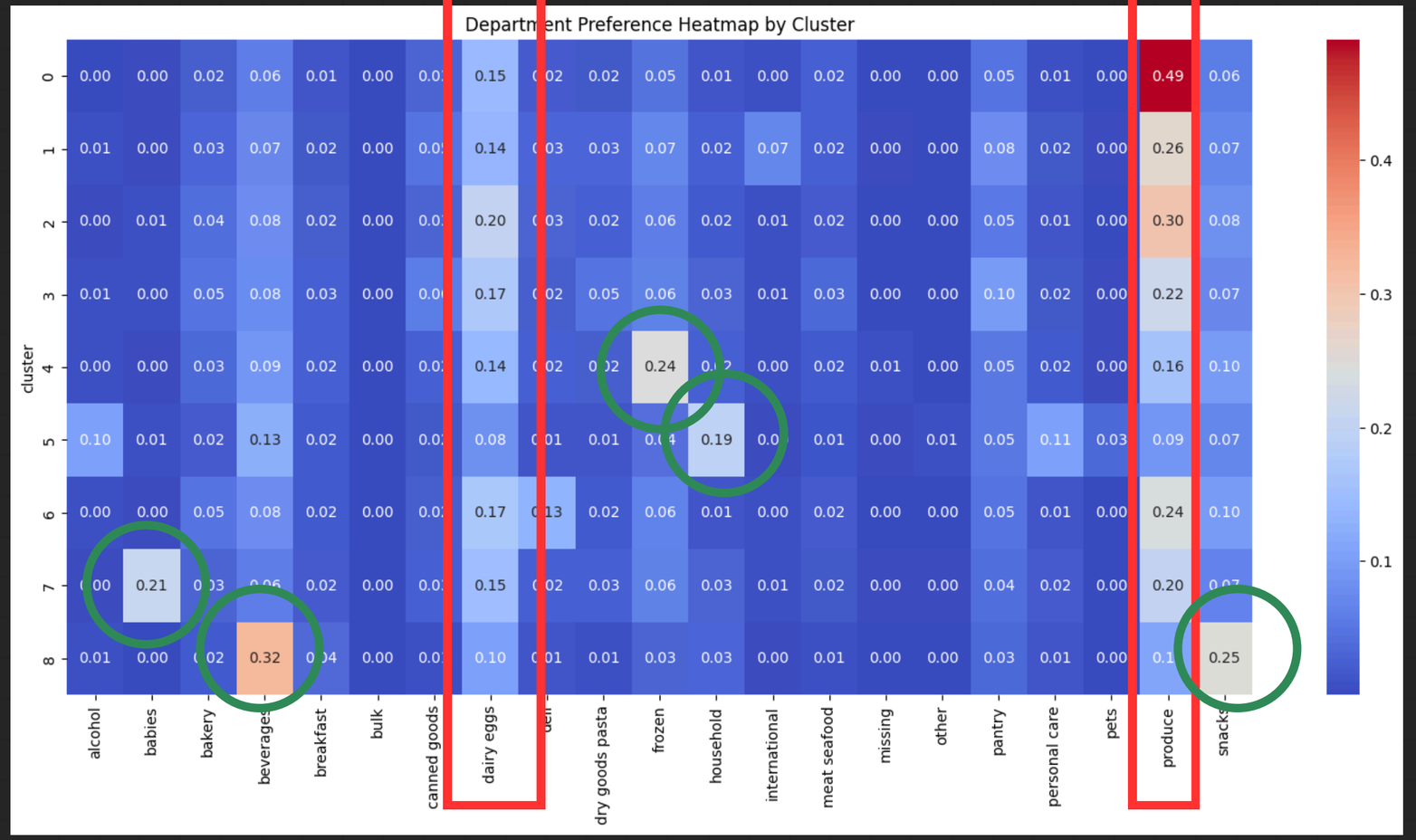


Figure 8. Highlighted Department Preference Patterns by Cluster

Behavioral and Product Preference Features

(Without Produce and Dairy Eggs)

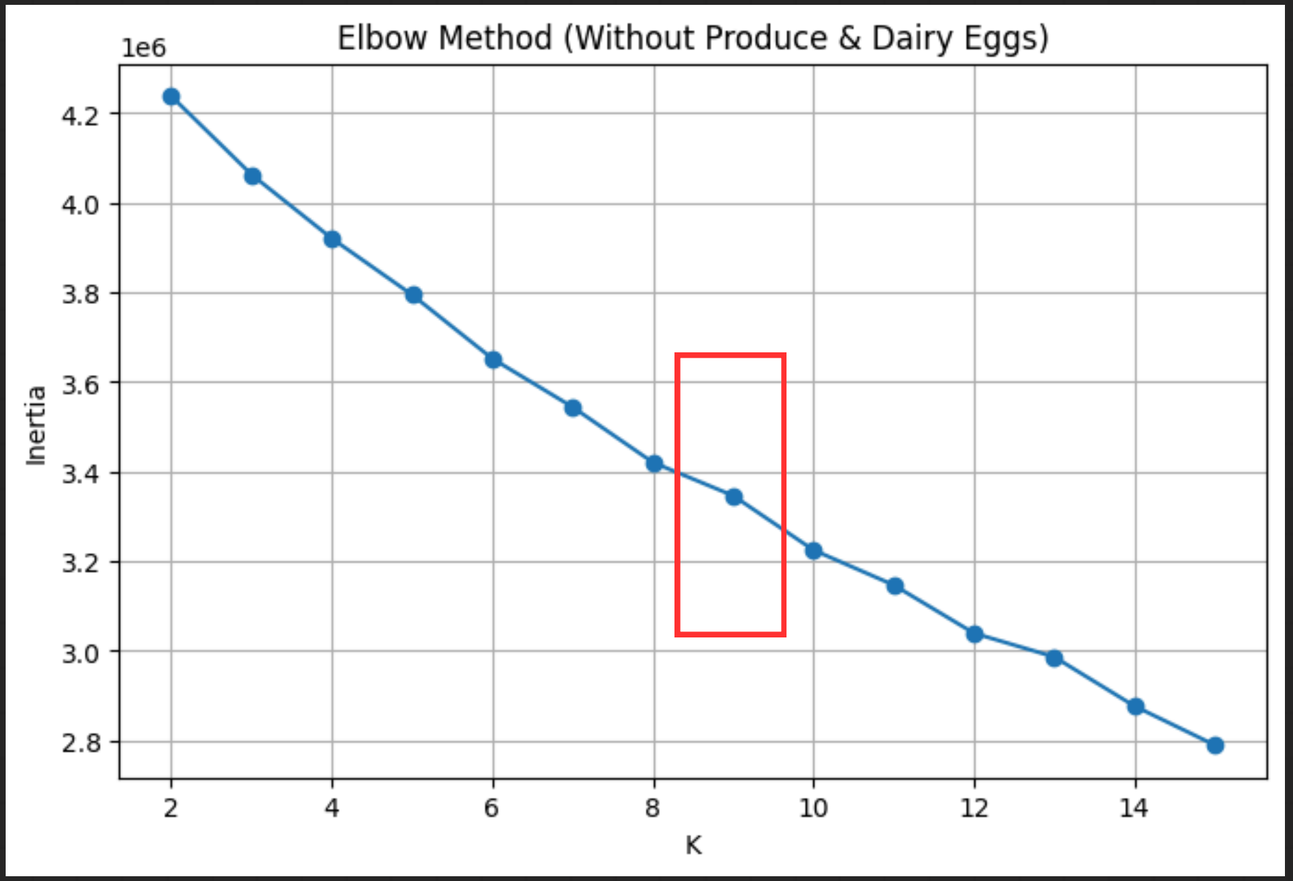


Figure 9. Elbow Method without Produce and Dairy Eggs



Figure 10. PCA Visualization without Produce and Dairy Eggs

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Behavioral and Product Preference Features ©

(Without Produce and Dairy Eggs)

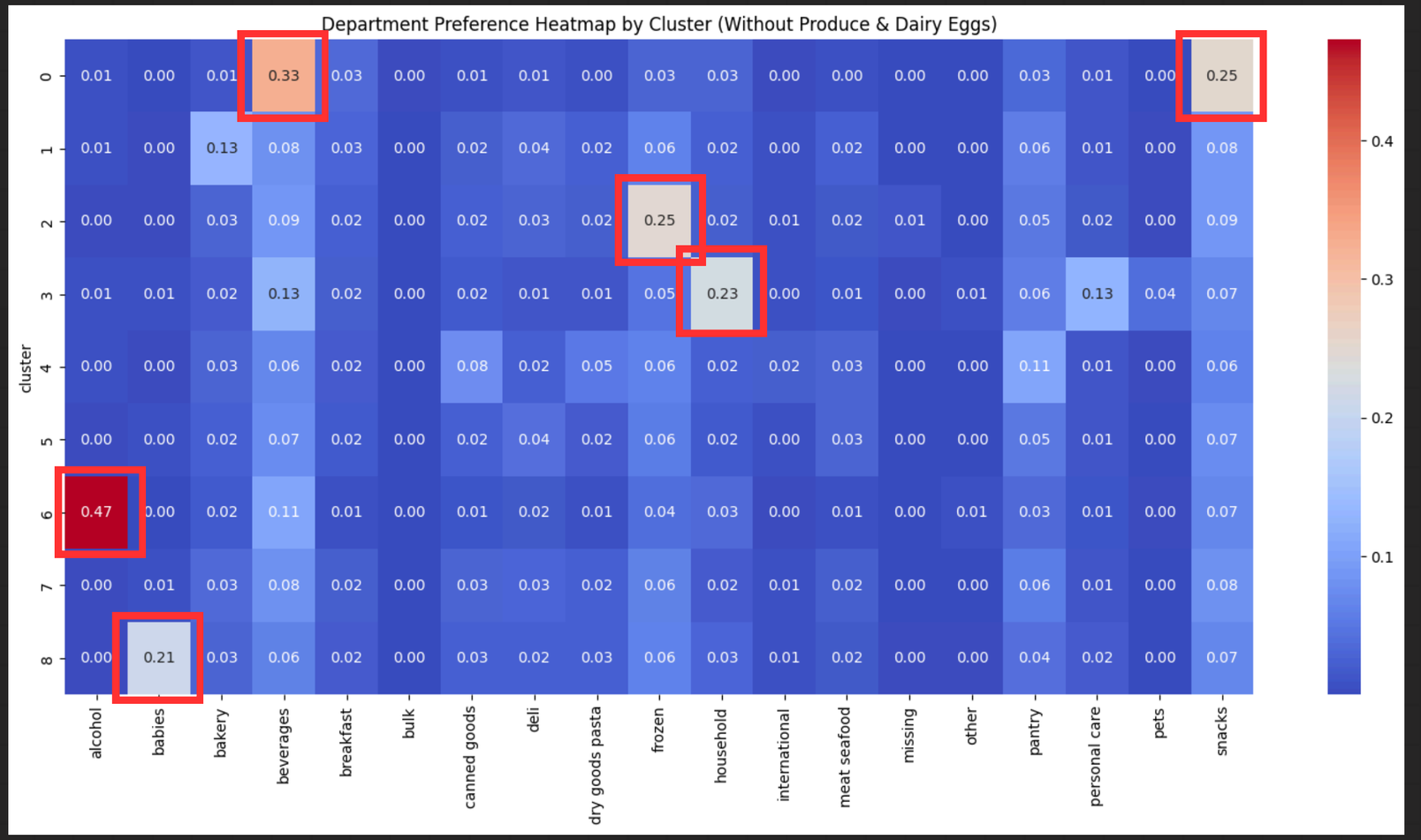


Figure 11. Department Preference Heatmap without Produce and Dairy Eggs

Behavioral and Product Preference Features©

(Without Produce and Dairy Eggs)

Specialized Customer Groups and Their Preferred Products:

Beverage & Snack Users (Cluster 0)

- Soda, Sparkling Water Grapefruit, Spring Water, Lime Sparkling Water.....

Household Users (Cluster 3)

- Ultra Soft Facial Tissues, White Giant Paper Towel Rolls, Select-A-Size White Paper Towels.....

Baby-product Users (Cluster 8)

- Baby Food Stage 2 Blueberry Pear & Purple Carrot, Spinach Peas & Pear Stage 2 Baby Food.....

Frozen-food Users (Cluster 2)

- Buleberries, Macaroni & Cheese, Organic Broccoli Florets, Chocolate Ice Cream.....

Alcohol-oriented Users (Cluster 6)

- India Pale Ale, Beer, Sauvignon Blanc, Cabernet Sauvignon, Chardonnay.....

Table 2. Specialized Customer Groups and Their Preferred Product

Behavioral and Product Preference Features ©

(Without Produce and Dairy Eggs)

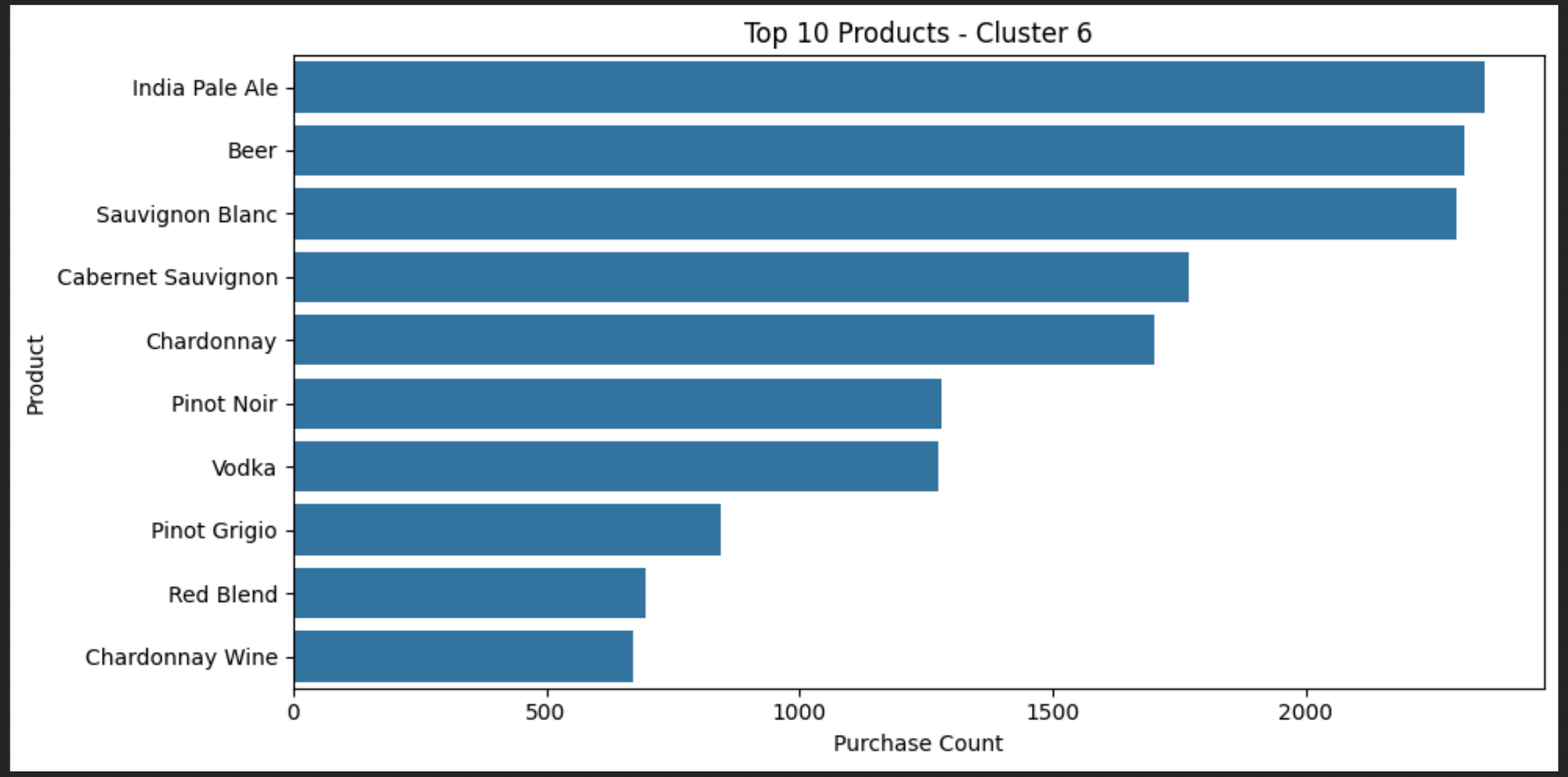


Figure 12. Top 10 Products of Alcohol-oriented Users

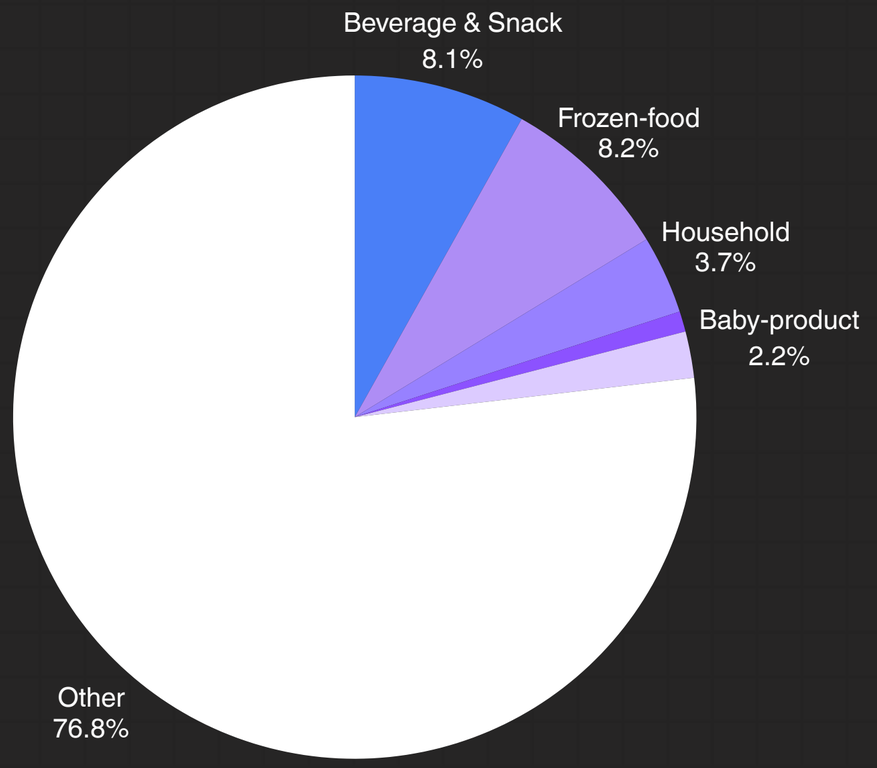
Behavioral and Product Preference Features ©

(Without Produce and Dairy Eggs)

Category	Number of User
Beverage & Snack	16753
Frozen-food	16842
Household	7647
Alcohol-oriented	1995
Baby-product	4521
General User	158451

Table 3. Number of Users by Product Preference Category

Figure 13. Distribution of Product Preference User Categories



Customer Behavioral and Product Preference Features ©

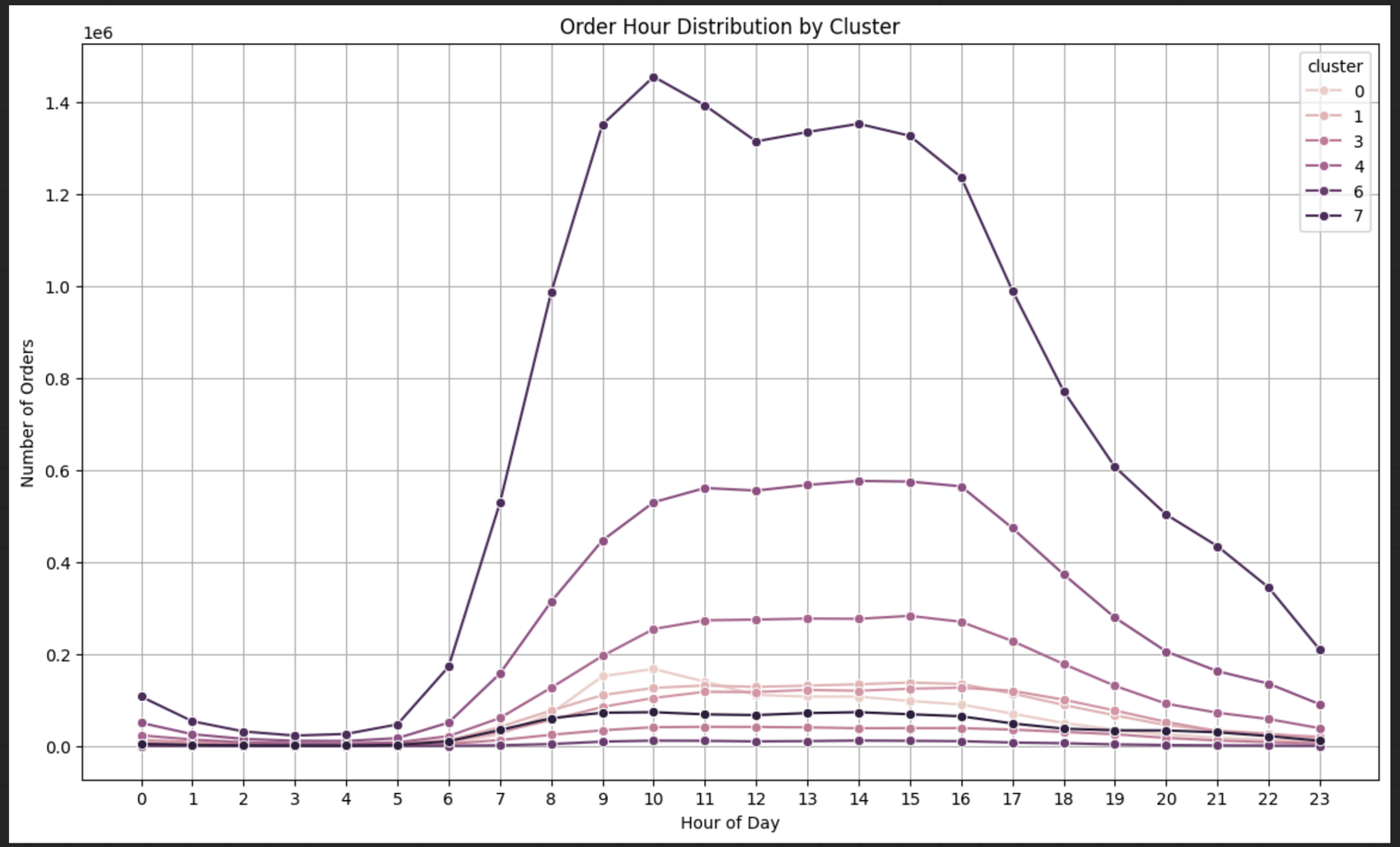


Figure 14. Order Hour Distribution by Customer Segment

Customer Behavioral and Product Preference Features ©

Beverage & Snack Users:

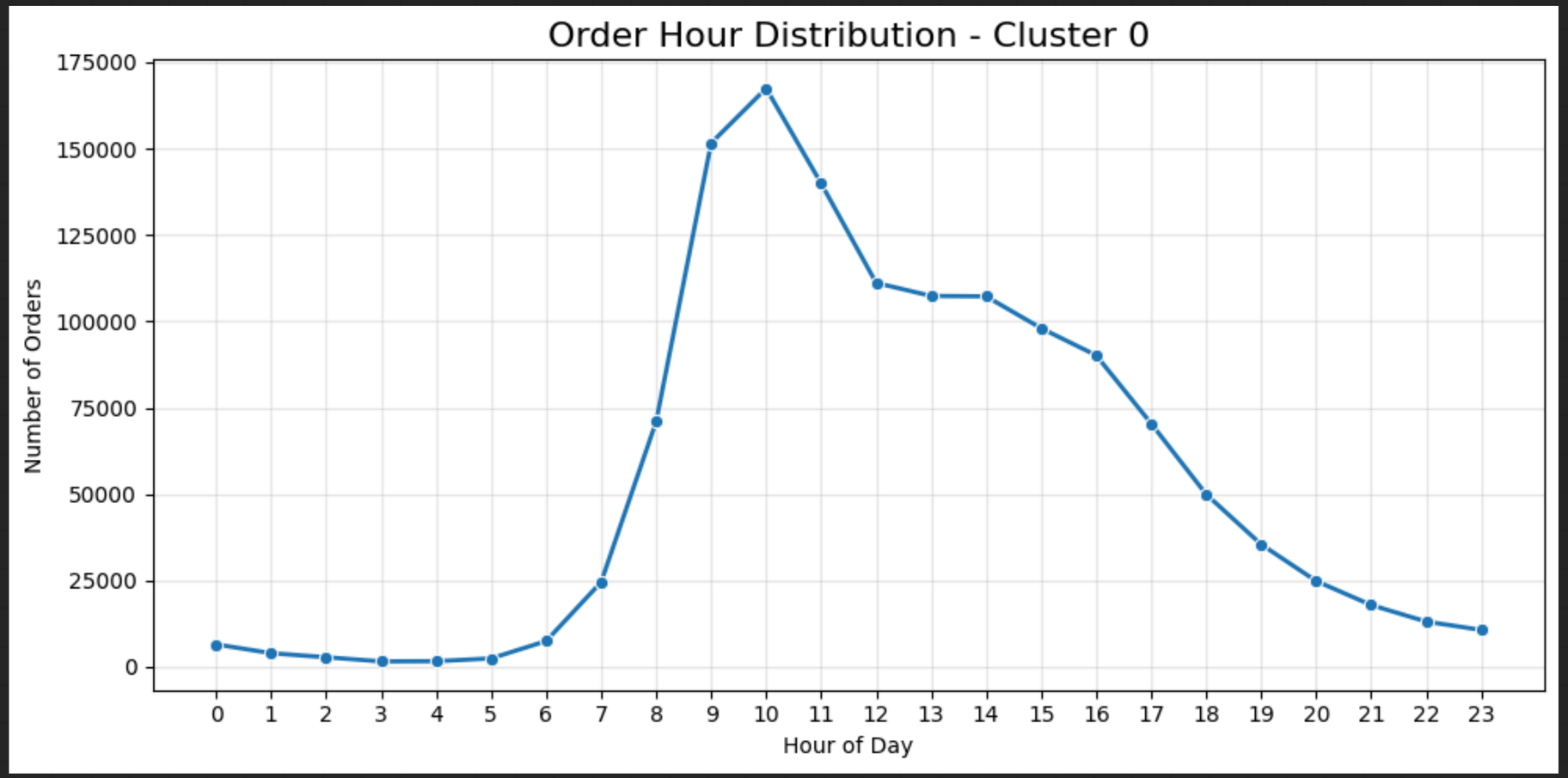


Figure 15. Order Hour Distribution of Beverage & Snack Users

EXPECT EXPERIMENTAL RESULTS

Basket Analysis

- Market Basket Analysis: Key Methods and Metrics
- Beverage & Snack Users: Basket Analysis Overview
- Basket Analysis Example for Alcohol-Oriented Users

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Market Basket Analysis: Key Methods and Metrics

We apply Market Basket Analysis to discover products that are frequently purchased together.

This helps us understand hidden purchasing relationships and supports recommendation system design.

Support

- Frequency of itemset in all transactions

$$Support(A, B) = \frac{\#\text{ transactions containing A and B}}{\text{total transactions}}$$

Confidence

- Probability of buying B given A

$$Confidence(A \rightarrow B) = \frac{Support(A, B)}{Support(A)}$$

Lift

- Strength of association between A and B

$$Lift(A \rightarrow B) = \frac{Confidence(A \rightarrow B)}{Support(B)}$$

Figure 16. Market Basket Analysis Metrics

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Beverage & Snack Users: Basket Analysis Overview

Basket Analysis Example

Beverage & Snack Users (Cluster 0)

Antecedents	Consequents	Support	Confidence	Lift
Lime Sparkling Water	Sparkling Water Grapefruit	0.0293	0.5487	7.0191
Sparkling Water Grapefruit	Lime Sparkling Water	0.0293	0.3750	7.0191
Sparkling Water Grapefruit	Sparkling Lemon Water	0.0248	0.3181	6.7003
Sparkling Lemon Water	Sparkling Water Grapefruit	0.0249	0.5238	6.7003

Table 4. Basket Analysis Rules for Beverage & Snack Users

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Basket Analysis Example for Alcohol-Oriented Users

Basket Analysis Example

Alcohol-oriented Users (Cluster 6)

Antecedents	Consequents	Support	Confidence	Lift
Beer	India Pale Ale	0.0422	0.3189	2.3681
India Pale Ale	Beer	0.0422	0.3136	2.3682
Sauvignon Blanc	Cabernet Sauvignon	0.0194	0.1475	1.4556
'Cabernet Sauvignon	Sauvignon Blanc	0.0194	0.1915	1.4556

Table 5. Basket Analysis Rules for Alcohol-oriented Users

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EXPECT EXPERIMENTAL RESULTS

Time-aware Analysis

- Time-Aware Basket Analysis: Motivation and Method Overview
- Beverage & Snack Users：Morning Example
- Beverage & Snack Users：Midnight Example

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Time-Aware Basket Analysis: Motivation and Method Overview

Customer purchasing behavior may change across different time periods.
Therefore, we further apply association rule mining under different time segments to discover time-dependent purchasing patterns.

This enables more context-aware recommendations, since products that are frequently purchased together in the morning may differ from those purchased together at night.

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Beverage & Snack Users：Morning Example

Time-aware Basket Analysis Example

Beverage & Snack Users (Cluster 0): Morning

Antecedents	Consequents	Support	Confidence	Lift
Lime Sparkling Water	Sparkling Water Grapefruit	0.0337	0.5662	6.5369
Sparkling Water Grapefruit	Lime Sparkling Water	0.0337	0.3893	6.5369
Sparkling Lemon Water	Sparkling Water Grapefruit	0.0286	0.5448	6.2892
Sparkling Water Grapefruit	Sparkling Lemon Water	0.0286	0.3299	6.2892

Table 6. Morning Time-aware Basket Analysis Rules for Beverage & Snack Users

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Beverage & Snack Users：Midnight Example

Time-aware Basket Analysis Example

Beverage & Snack Users (Cluster 0): Midnight

Antecedents	Consequents	Support	Confidence	Lift
Sparkling Water Grapefruit	Lime Sparkling Water	0.0164	0.3249	9.5887
Lime Sparkling Water	Sparkling Water Grapefruit	0.0164	0.4848	9.5887
Lime Sparkling Water	Sparkling Lemon Water	0.0136	0.4015	12.3173
Sparkling Lemon Water	Lime Sparkling Water	0.0136	0.4173	12.3173

Table 7. Midnight Time-aware Basket Analysis Rules for Beverage & Snack Users

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5 CONCLUSION AND ANALYSIS ©

Customer Segmentation

- Behavioral features help identify customers with different loyalty levels.
- Combining behavioral and product preference features reveals specialized customer groups with distinct purchasing preferences.
- These insights enable businesses to develop targeted marketing strategies for different customer segments.

Market Basket Analysis

- Association rule mining reveals products that are frequently purchased together within different customer segments.
- These purchasing relationships can support personalized product recommendations and cross-selling strategies.

Time-aware Basket Analysis

- Product associations vary across different time periods.
- Incorporating time information allows businesses to develop more context-aware recommendation strategies.

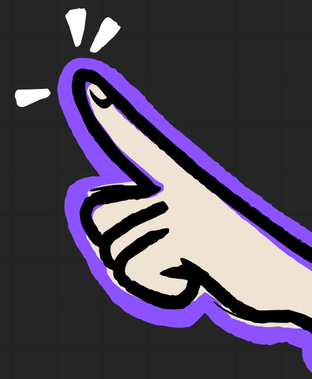
THANK *YOU*

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Final Project

Group I

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Q&A

